

Pen-DHRL-TD: Teacher-Distilled Hierarchical RL for Autonomous Penetration Testing

NASimEmu-enhanced Project

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Summary. Pen-DHRL-TD augments a deep hierarchical RL pentesting agent (NASimNetDHRL: a recurrent GNN *manager* + goal-conditioned graph-attention *worker*, 212k params) with offline supervision from a locally-hosted LLM teacher (qwen2.5:3b-instruct via Ollama, never queried inside the training loop). The manager selects one of 8 named semantic subgoals (e.g. GAIN_INITIAL_ACCESS, ESCALATE_PRIVILEGE); the teacher labels sanitized, partially-observable state summaries with a goal + confidence, producing a provenance-tracked dataset distilled into the subgoal head via a confidence-weighted cross-entropy loss, folded into a four-stage schedule (supervised warm-start → decaying joint PPO+distillation → guaranteed RL-only tail). A live *explainability monitor* prints the manager’s realized subgoal histogram and distillation diagnostics every epoch.

Method highlights.

- **Teacher pipeline:** discovered-only state summary → JSON-schema-constrained LLM query → anti-hallucination/executable-text validator → retry-on-reject, never silently discarded.
- **Triggered sampling:** teacher queried only at informative moments (new subnet, first compromise, privilege change, IDS alert, repeated failure, high manager entropy, periodic fallback) under a diversified rollout-policy mixture.
- **Distillation loss:** $\mathcal{L}_{\text{goal}} = \text{mean}_i [\text{clip}(c_i, 0.05, 1) \cdot (-\log \hat{p}_\theta(g_i | s_i))]$, jointly trained with PPO under an annealed weight, forced to exactly 0 for the final 10% of training.

Key findings (12-subnet, 72–96-host dynamic scenario family; run b33undf9, 200 epochs):

- **Teacher dataset:** 2,235 labeled states, 89.5% valid, but severely class-imbalanced (84% DISCOVER_SUBNET; RECOVER_OR_REPLAN: 0 examples), which the live subgoal histogram traced directly to a training-time subgoal collapse (92%→100% DISCOVER_SUBNET by training’s end).
- **Checkpoint-selection pitfall:** the default “best-checkpoint” metric (`captured_avg`) is blind to episode length and froze on a mid-run checkpoint, missing a large late-training efficiency gain visible only in `reward_avg_episodes`.
- **Outperforms a non-distilled Pen-DHRL baseline:** same architecture, same parameter count (212k), trained *without* the teacher signal. The distilled agent wins on every metric, every scenario.

Scenario	reward_avg	reward_avg_episodes	eplen_avg	captured_avg
<i>Pen-DHRL-TD (final checkpoint, LLM-distilled)</i>				
base	2.3216	259.12	111.61	29.63
varA	0.2967	80.47	271.18	17.50
varB	0.1688	46.04	272.69	13.25
bridge	1.7425	205.48	117.93	24.67
<i>Pen-DHRL baseline (same architecture/params, no distillation)</i>				
base	0.5428	217.11	400.0	29.59
varA	0.1708	68.33	400.0	13.73
varB	0.0778	31.12	400.0	9.39
bridge	1.1470	161.96	141.20	23.36

Table 1: 100 held-out episodes/scenario. Baseline: stronger of two independently trained runs per scenario.

Takeaway. Pen-DHRL-TD wins on every metric on every scenario, and terminates well before the 400-step limit on all four (the baseline only does so on **bridge**). Because the two runs share architecture and parameter count exactly, the gain traces to the LLM-teacher distillation signal itself, not extra model capacity.

Full technical report (18 pp.: architecture, teacher pipeline, 5 algorithms, complete results and limitations) available on request.